3.8 Graphing Absolute Value Functions

Name _____ Period ____ Date _____

Algebra I 3.8 · Student sheet

I can…

- graph absolute value functions and find their range.
- describe and write transformations of f(x) = |x|.
- find the vertex of an absolute value function.

Vocabulary · fill in the missing word

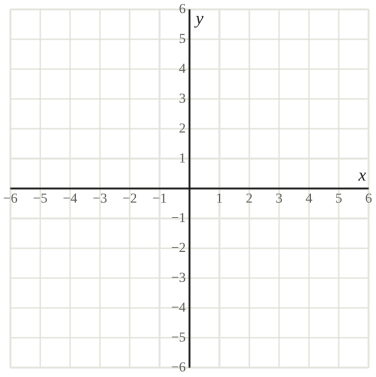
Absolute value function — f(x) = |x|, a _____ graph with vertex (0, 0)

Vertex — the point where the graph changes _____

g(x) = a|x − h| + k — _____ (h, k); a < 0 opens down

Part 1 · Make the V

x	−3	−2	−1	0	1	2	3
f(x) = |x|							
g(x) = |x − 2| + 1							



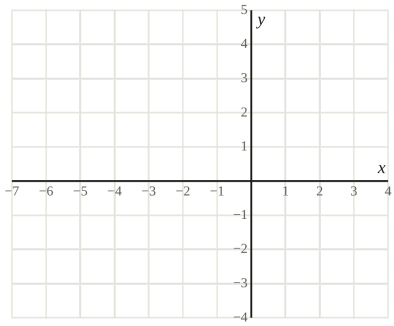
Vertex of f: _____ Vertex of g: _____ How did g move? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Vertex: g(x) = a|x − h| + k has vertex _____.

Example 1

Graph g(x) = |x + 2| − 3. Give the vertex and the range.



Opening: a < 0: the graph opens _____.

Stretch or shrink: |a| > 1: the graph is _____; 0 < |a| < 1: it is _____.

Example 2

Find the vertex of h(x) = −2|x − 1| + 4. Does it open up or down? Give the range.

Example 3

Write a function g whose graph is the graph of f(x) = |x| moved 3 units right and 2 units up.

Part 3 · Partner practice

Partner 1

Find the vertex of y = |x − 5| − 1.

Partner 2

Find the range of $y = 3|x| + 2$.